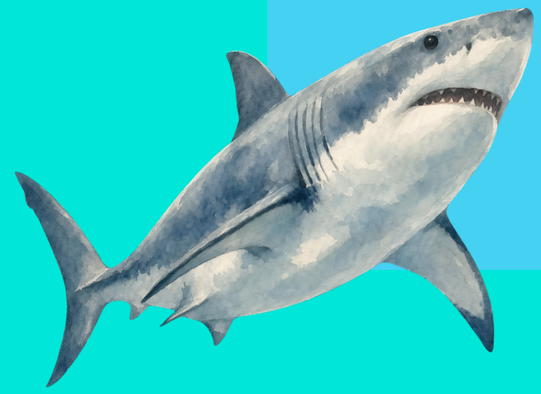
SHARK ATTACK DATA ANALYSIS

Findings and recommendations for insurance pricing and coverage



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IRON HACK

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Project objective

We are the data analytics team of an **insurance company**.

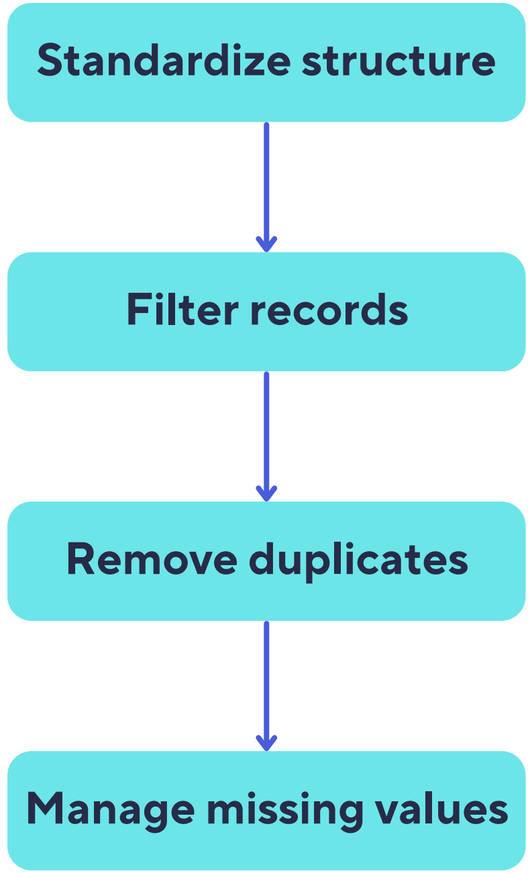
We’re analyzing shark incident records to better understand patterns and use these findings to guide pricing and coverage decisions.

Our goals:

- Identify patterns in recorded attacks
- Test our assumptions against the data
- Inform pricing and coverage recommendations

Data cleaning & transformation

- Clean column names to apply consistent naming conventions.
- Convert dates, numbers and text to appropriate formats.
- Filter records to only retain records from 2000 onward.
- Address invalid and missing values.
- Identify and remove duplicate records.



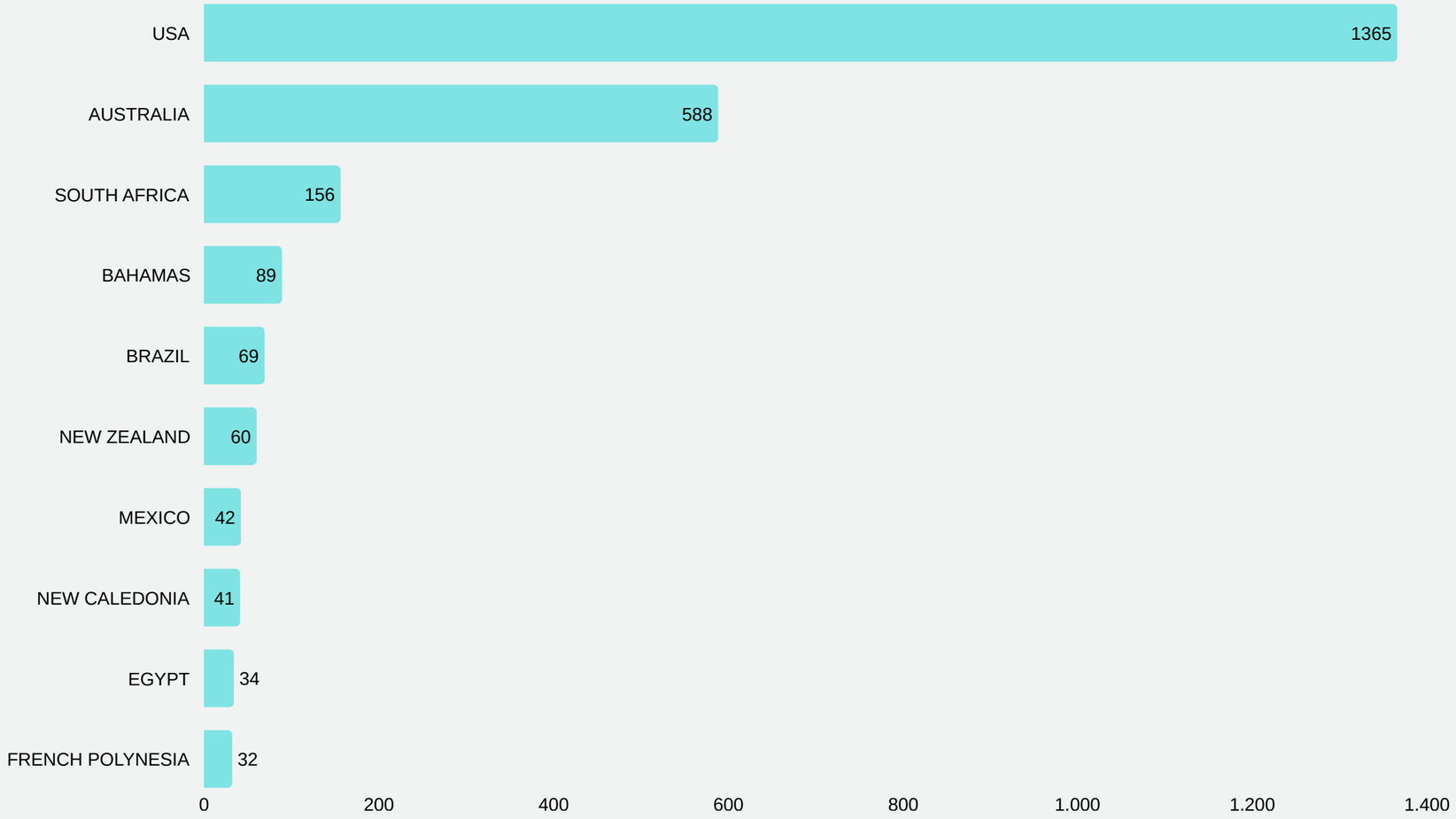
H1: Men are more likely to suffer shark attacks



4 out of 5 attacks happen to men

H2: USA and Australia lead in shark attacks

Number of attacks in the 10 countries with the most attacks



55% of top-10 attacks are in the USA

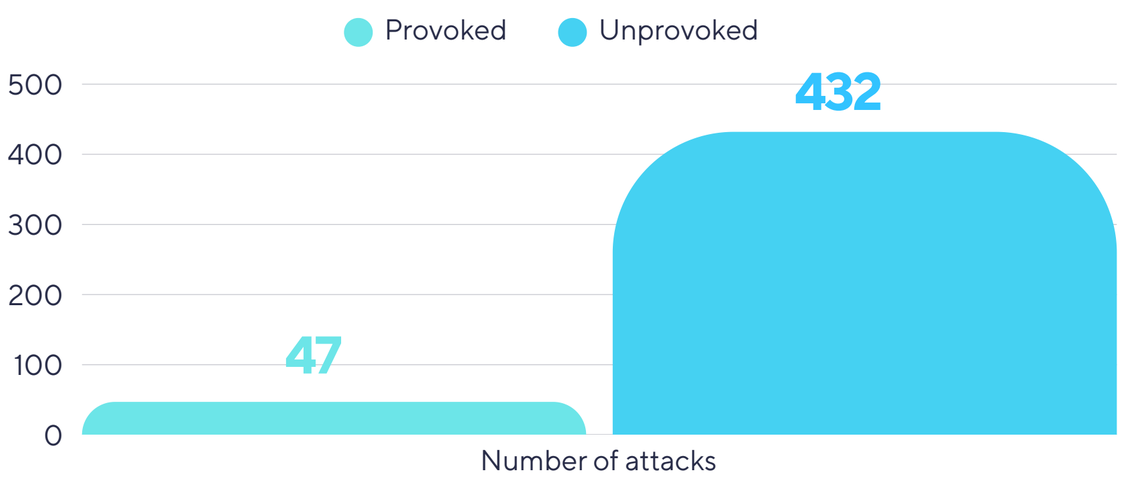
79% are in the USA and Australia combined

Conclusion: Attacks are concentrated in a few countries. The USA and Australia stand out, with South Africa a distant third; every other country is under 4%.

Percentages are out of 2,476 attacks (total of these 10 countries).

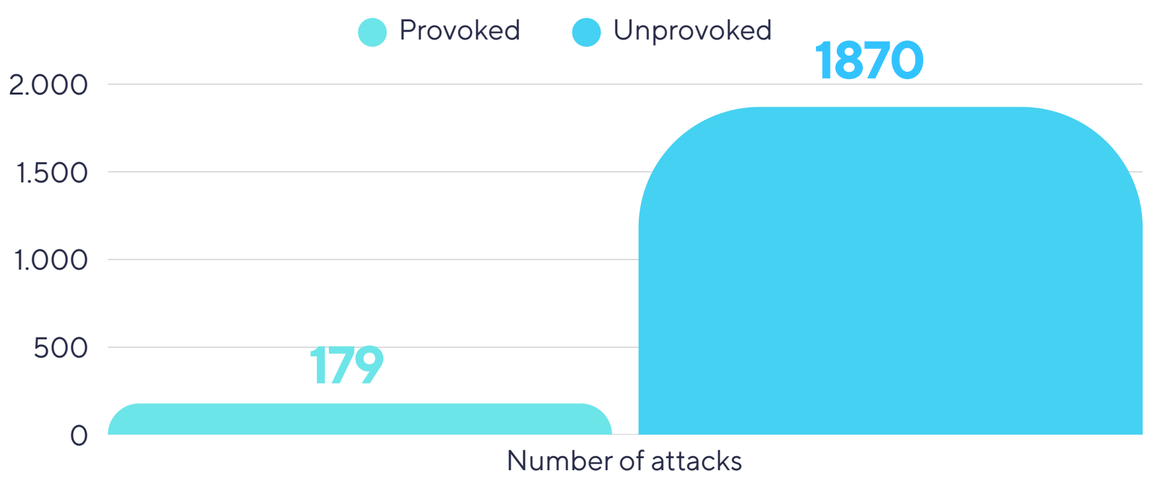
H3: Men are more likely to provoke shark attacks than women

Women



- **About 10% of attacks on women were provoked.**

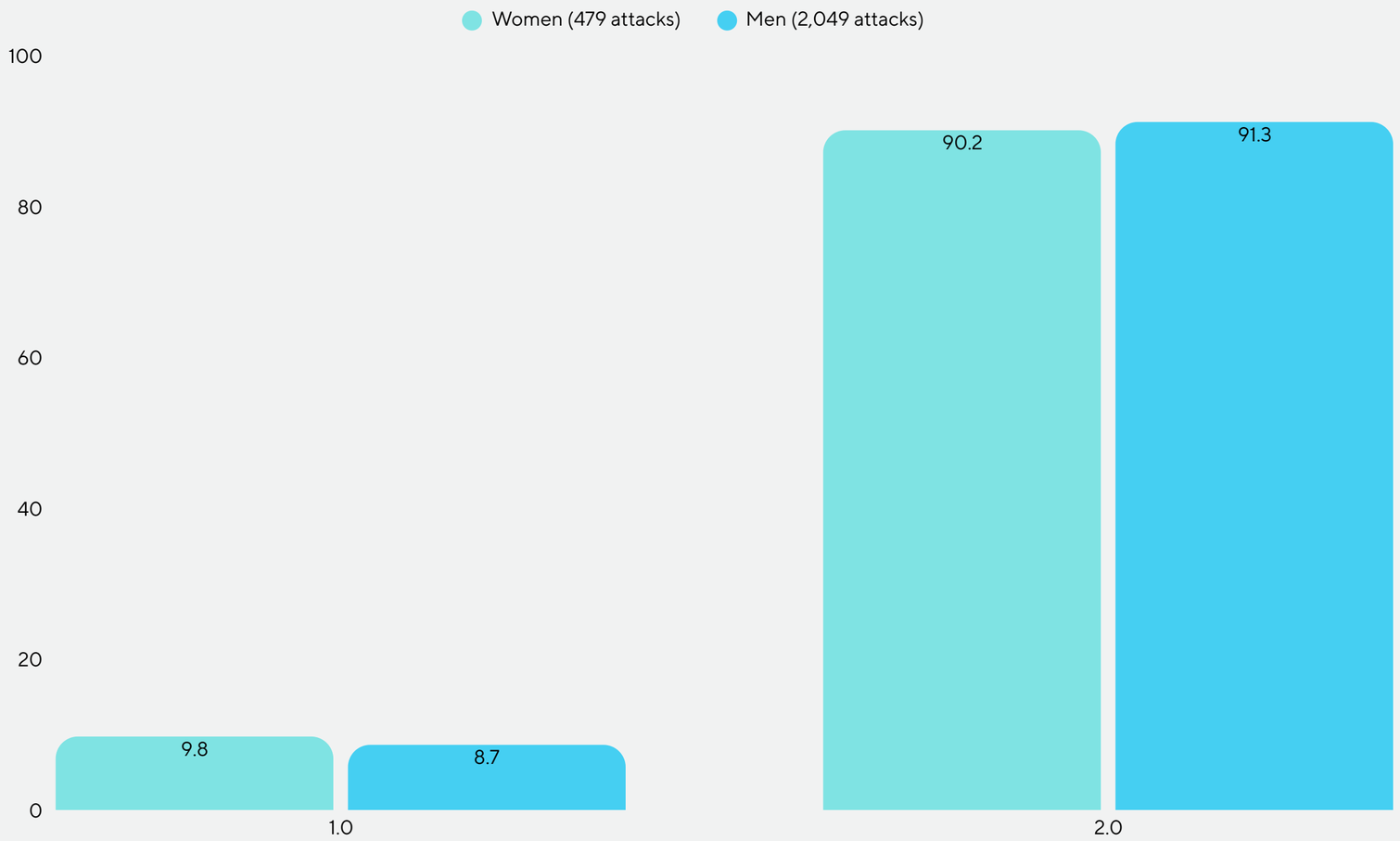
Men



- **About 9.5% of attacks on men were provoked.**

Over 90% of attacks are unprovoked

Share of attacks by type, women vs. men (%)



9.8% of attacks on women were provoked

8.7% of attacks on men were provoked

Conclusion: Over 9 in 10 attacks are unprovoked for both women and men. The difference between the sexes is only about 1 point, so sex has almost no effect on whether an attack is provoked.

Business recommendations

Based on our initial analysis, we suggest the following:

Consider higher premiums for men	Larger share of recorded incidents warrants reviewing gender differences in pricing.
Adjust premiums based on location	Pricing should be re-evaluated for USA and Australia, where most attacks have been recorded.
Ensure consistent provocation coverage	Apply the same provocation coverage conditions across genders.

Next steps

Continue analyzing the dataset to **identify other possible patterns**, and make recommendations accordingly.

- **Activity-based pricing:** Compare attacks across activities (surfing, swimming, diving, ...) to validate whether an activity-based pricing should be implemented.
- **Seasonal pricing:** Explore whether attacks peak during particular months or seasons of the year.
- **Coverage limitations:** Examine injury severity across activities.

THANK YOU